



Final Exam (60 Points) - Model (1)

Q1: [Multiple Choice] Identify the letter of the choice that best answers the question

[36 x 1.0 = 36 points]

1. Set of wires that acts as a shared data path to connect multiple subsystems within the system is ____
~~(A)~~ system bus (B) I/O subsystem (C) Bus (D) Control bus
2. ____ allow the devices themselves to determine who has the highest priority and get the bus.
(A) Distributed arbitration using collision detection (B) Distributed arbitration using self-selection ~~(C)~~ Centralized parallel arbitration (D) Daisy chain arbitration
3. Getting the data from address location to be placed on the data bus refers to ____
(A) I/O write (B) I/O read (C) memory write ~~(D)~~ Memory read
4. ____ provides an efficient mode of communication between the central system and the outside environment.
(A) Processor ~~(B)~~ I/O subsystem (C) Bus (D) Memory
5. ____ is built into the chassis of the machine and connects the processor, I/O devices, and memory (All devices share one bus).
(A) I/O Bus (B) Processor-Memory bus ~~(C)~~ Backplane bus (D) System bus
6. ____ is a unidirectional bus.
~~(A)~~ address bus (B) data bus (C) control bus (D) system bus
7. ____ can be assigned to a set of functions by the programmer. Sometimes their use within instruction set is orthogonal to operation.
~~(A)~~ General purpose register (B) Data register (C) Address register (D) Condition codes register
8. ____ contains a word of data to be written to memory or the word most recently read.
(A) Program counter (B) Instruction register (C) Memory address register ~~(D)~~ Memory buffer register
9. In ____, the execution of an instruction may require reading data from memory or an I/O module.
(A) Fetch instruction (B) Interpret instruction ~~(C)~~ Fetch data (D) Process data
10. ____ means doing more of work in the same time.
(A) Performance ~~(B)~~ Throughput (C) Availability (D) Scalability
- ~~11.~~ Many processors + distributed memory + communication via network (messages) results to ____ *mep*
(A) Symmetric MultiProcessors ~~(B)~~ Massively Parallel Processors (C) Multiple Instruction, Single Data (D) Single Instruction, Multiple Data
- ~~12.~~ Master / worker cluster, map / reduce framework are examples of ____
(A) Single Program Multiple Data *smp* (B) Multiple Program Multiple Data *mmp* (C) Single instruction, single data *sisd* ~~(D)~~ Multiple instruction, single data *misd*
13. ____ has equal load and store access for all processors to all memory.
~~(A)~~ Uniform memory access (B) Non-uniform memory access (C) Distributed Memory (D) Hybrid
14. ____ used by control unit to control operation of processor and operating system programs to control the execution of programs.
(A) User-visible registers ~~(B)~~ Control and status registers (C) Data registers (D) Address registers
15. ____ means dividing processing of a CPU instruction into a series of independent steps of microinstructions.
~~(A)~~ Pipelining (B) Instruction-level parallelism (C) Branch prediction (D) Delayed Branch
16. ____ occurs when pipeline makes wrong decision on a branch prediction and therefore the pipeline must subsequently be discarded.
(A) Resource hazard (B) Data hazard ~~(C)~~ control hazard (D) Instruction hazard
17. ____ contains the address of a location in memory.
(A) Memory buffer register (B) Instruction register (C) Program counter ~~(D)~~ Memory address register
- ~~18.~~ Mechanism by which other modules (I/O, memory) can cut off the normal processing of processor, is ____
(A) Pipelining (B) Resource hazard ~~(C)~~ Interrupt (D) Distributed arbitration using collision detection
19. ____ generated by condition that occurs as a result of an instruction execution, such as arithmetic overflow or division by zero.
~~(A)~~ Program interrupt (B) Timer Interrupt (C) I/O interrupt (D) Hardware interrupt
- ~~20.~~ In ____, processors share access to main memory over one bus.
(A) clusters (B) massively parallel processors (C) symmetric Multiprocessing (D) asymmetric multiprocessing
21. Architecture describes how the computer do its operations. *h n p*
(A) TRUE ~~(B)~~ FALSE *as n p*
22. Intel and AMD does not use the same architecture.
(A) TRUE ~~(B)~~ FALSE

- Control memory stores microprogram.
- (A) TRUE (B) FALSE
- In cache memory scheme, level n is smaller and faster than level n + 1.
- TRUE (B) FALSE
- Register that refers to the next instruction to be fetched is —
- (A) MBR (B) MAR (C) PC (D) IR
- Fixed point representation of real number has a specific number of digits reserved for the integer part and fractional part.
- TRUE (B) FALSE
- Sign-magnitude representation of "10000001" refers to decimal value —
- (A) -1 (B) 65 (C) -126 (D) -127
- Overflow can be detected by XOR the two bits of carry-in and carry-out.
- TRUE (B) FALSE
- In IEEE 64-bit representations, the fractional part of the mantissa occupies — bits.
- (A) 8 (B) 23 (C) 11 (D) 52
- In IEEE single-precision representations, the exponent part occupies — bits.
- (A) 8 (B) 23 (C) 11 (D) 52
- ASCII code uses — bits to represent each character.
- (A) 7 (B) 8 (C) 16 (D) 32
- UTF-8 is a variable width character encoding while Unicode is fixed width character encoding.
- TRUE (B) FALSE
- Big-endian system has property that same value can be read from memory at different lengths without using different address.
- (A) TRUE (B) FALSE
- A processor can simultaneously work on multiple instructions, this refers to —
- (A) Branch prediction (B) Pipelining (C) Data flow analysis (D) Speculative execution
- RISC systems improve performance by reducing the number of instructions per program.
- TRUE (B) FALSE
- ARM architecture refers to a processor architecture that has evolved from RISC.
- (A) TRUE (B) FALSE

22: [Multiple Choice] Identify the letter of the choice that best answers the question

1. In IEEE-754 32-bit floating-point, hexadecimal value "3FC0-0000" represent decimal value —
- (A) -1 (B) 1 (C) 1.5 (D) -1.5
2. Convert decimal number 3.5 into IEEE-754 32-bit floating-point representation is —
- (A) 4060-0000H (B) C060-0000H (C) 4000-0000H (D) C000-0000H
3. The Average memory access time can be improved by —
- (A) increasing miss rate (B) reducing hit rate (C) decreasing miss rate (D) reducing the memory size
4. All the following is True about DRAM except —
- (A) It requires supporting refresh circuitry. (B) It is less dense than SRAM. (C) It is a volatile memory.
5. Suppose a two levels memory organization, What is the number of data values supplied to the processor by main memory is 7 % and total number of data access instructions is 4000.
- (A) 280 (B) 3720 (C) 2800 (D) 1200
6. Consider a direct mapping with a total of 8 blocks. How many victim block we need to replace for the following addresses 8,12,0,3,6,9,5,8 ?
- (A) 2 (B) 4 (C) 6 (D) 8

Consider a 2-way set associative mapping with 8 cache blocks. The main memory consists of 128 blocks and each block contains 4 words. The memory block requests are in the following order: 0,5,3,9,7,0,16,55

7. If the FIFO replacement policy is applied, How many cache misses? (Assuming that initially the cache is empty).
- (A) 4 (B) 8 (C) 1 (D) 7
8. What memory blocks will be present in the cache after if LRU cache replacement policy is applied. (Assuming, initially the cache is empty)
- (A) 0,3,5,7,16,55 (B) 0,3,5,7,9,16,55 (C) 0,5,7,9,16,55 (D) 3,5,7,9,16,55
9. The physical address is represented in —
- (A) 6 bits (B) 7 bits (C) 9 bits (D) 13 bits
10. The number of bits in tag index = —
- (A) 5 (B) 6 (C) 7 (D) 13
11. The number of sets in cache = —
- (A) 4 (B) 2 (C) 3 (D) 8
12. The number of bits in Block offset = —
- (A) 2 (B) 4 (C) 6 (D) 8

اللهم صلي وسلم وبارك
علي سيدنا محمد و علي
ال سيدنا محمد

Dr. Muhammad Haggag